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BIO 539: Final Project

Size Variation and Environmental Drivers of Sea Urchin Biomass

Introduction:

Sea urchins are important grazers in marine benthic ecosystems and can strongly influence the balance between macro-algae dominated habitats (kelp) and the creation of algal turf dominated habitats (barrens). Understanding the variation of sea urchin biomass is critical to understand how different species have preferences in habitat and diet.

Biomass is useful in measuring individual condition of urchins since it reflects the size and possible energetic/gonadal reserves. Test diameter is a standard measuring technique for sea urchin growth and should be positivity associated with an increase in biomass; however, biomass may vary with species, habitat, growth stages, and temporally. Beyond the sea urchin size, other factors can influence biomass. Species differences may reflect physiological variation, and habitat differences (kelp vs barren habitats) may reflect differences in food availability. Additionally, temporal variation may reflect environmental change between sampling years.

This project addresses two research questions:

1. What is the relationship between sea urchin diameter and biomass, and does this relationship differ between species?
2. After accounting for body size, how do species, habitat, and year influence variation in biomass?

It is hypothesized that biomass will increase with diameter, the scaling relationship will differ between the two species, and individuals in the kelp habitat will show higher biomass than those in the barren habitats, with year to having a small effect.

Methods:

Data Collection:

The data set consists of 2,287 sea urchin observations collected across two years, 2022 and 2023. Each observation includes habitat type (barren or kelp), quadrat number, species identity, color description, diameter (cm), and wet biomass (g). the two species observed were: *Strongylocentrotus droebachiensis* (n = 1,986) and *Strongylocentrotus pallidus* (n = 301). Most samples were collected from barren habitats (n = 1,988) and fewer from kelp habitats (n = 299). Data was from a collection by Koch, et al., 2024, that was posted on PANGAEA Data Publisher.

Citation:
Koch, Marie; Sickert, Chiara; Saborowski, Reinhard; Jungblut, Simon (2024): Sea urchin population in the Arctic Porsangerfjord [dataset]. PANGAEA, https://doi.org/10.1594/PANGAEA.964580

Data Preparation:

Data was downloaded from the above citation, opened a s .tab file, converted to a .txt and imported to excel to examine. I removed irrelevant columns: species unique identification, species unique identification (semantic URI), species unique identification (URI), and comments. I then converted it "From Text/CSV" to download a .csv file which was then imported into R Studio (Posit) and cleaned names prior to analysis. Categorical variables (species, habitat, year, and color) were converted to factors, and a small number of missing values were identified (1 for diameter and 4 for biomass) and these observations were excluded automatically during analysis. A Github repository was then created and connected to my R Studio project on Posit.

Exploratory Analysis:

Summary statistics and visualizations were used to explore the data. Scatter plots were used to examine the relationship between diameter and biomass, log-log plots were used to evaluate allometric scaling, and boxplots were used to compare biomass across species, habitat, and year. These visualizations guided the selection of statistical models I used.

Statistical Analysis:

Research Question 1: A linear model was used to test whether the relationship between diameter and biomass differed between species. I used both linear models (biomass ~ diameter and species) and log (allometric) models (log: biomass ~log: diameter and species) to see the comparison.

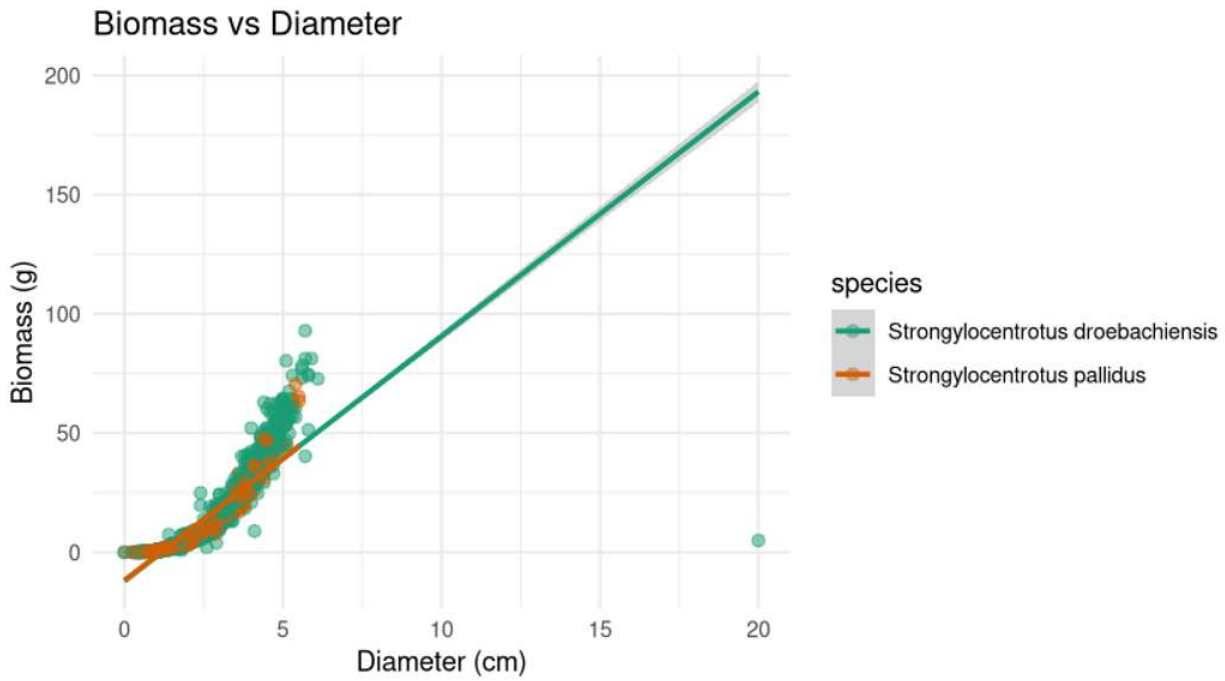
Research Question 2: A multiple regression model (biomass ~ diameter, species, habitat, and year) was used to evaluate the influence of ecological factors while controlling for size, and the model was assessed using R^2 and statistical significance using p-values.

Results:

A visual analysis of graphs and models shows a strong positive relationship between diameter and biomass (Figure 1: A, B). Larger sea urchins consistently had greater biomass, and this relationship appeared approximately linear across the observed range.

Diameter was eh dominate predictor of biomass ($p < 2 \times 10^{-16}$) which helped to explain the variation of ($R^2 = 0.776$). Species and diameter interactions were not statistically significant.

A:



B:

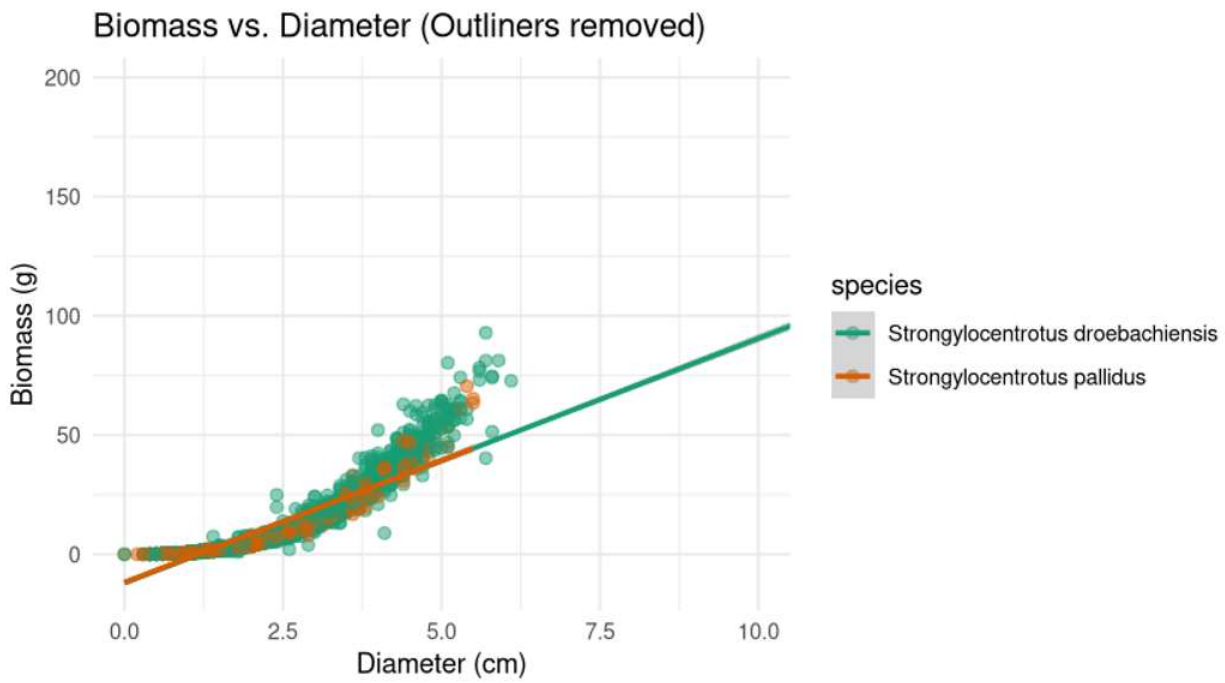


Figure 1 (A, B). Relationship between sea urchin diameter (cm) and biomass (g) for both species. Points represent individual observations and lines shown are fitted with linear regressions for each species. A: includes outlines B: Excluded outliners for closer look at relationships. Biomass increases strongly with diameter, indicating that body size is a major factor in biomass. Differences between species are minimal in these linear representations.

Log–log plots suggested an approximately linear relationship (Figure 2), which is consistent with allometric scaling in other species. The interaction between diameter and biomass was significant ($p < 2 \times 10^{-16}$), which indicated that biomass scales nonlinearly with size following that allometric relationship. Species differences were not significantly related (p=0.109), but the interactions between species and log diameter were statistically significant (p=0.0445). This suggests a small but noticeable difference in scaling relationships between species, with biomass increasing at slightly different rates with size.

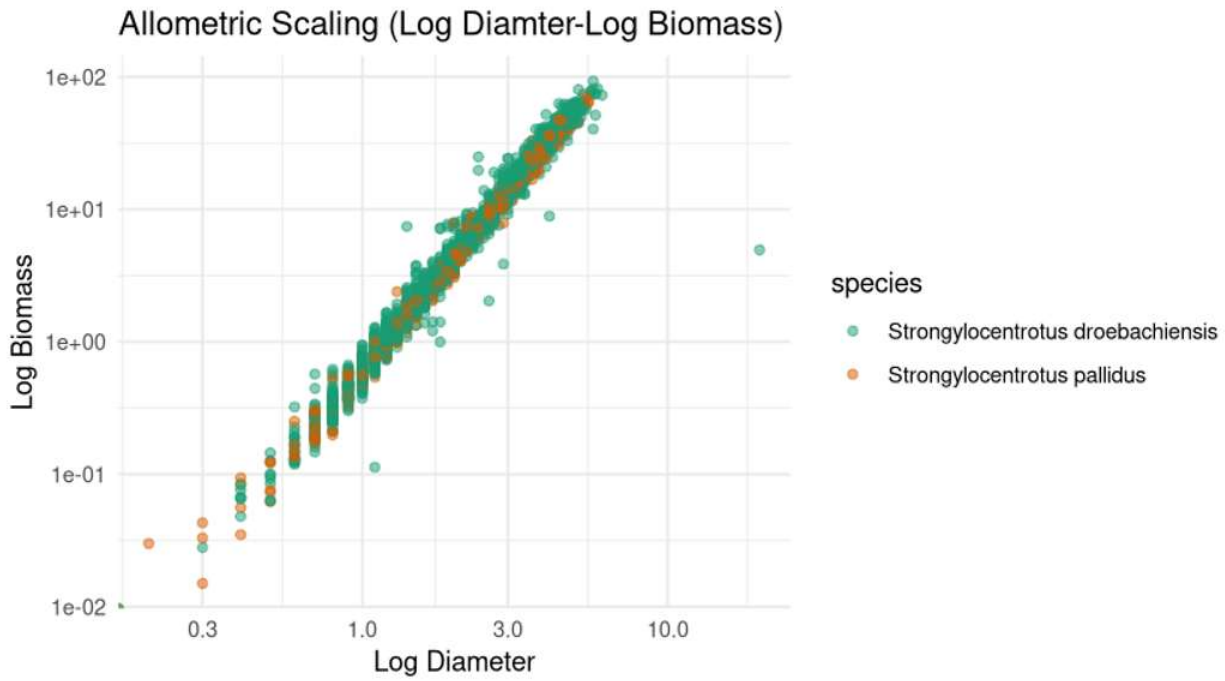


Figure 2. Log–log relationship between diameter and biomass for sea urchins. Each point represents an individual, and both axes are $\log_{10}$-transformed. The linear trend in log space indicates an allometric scaling relationship, with biomass increasing nonlinearly with size. Slight differences in slope between species reflect small but significant differences in scaling relationships.

Box plots showed that biomass distributions differed between species, habitats, and years (Figure 3). Urchins in kelp habitats generally exhibited higher biomass values than those in barren habitats ($p < 2 \times 10^{-16}$). Differences between years were less pronounced but still visible.

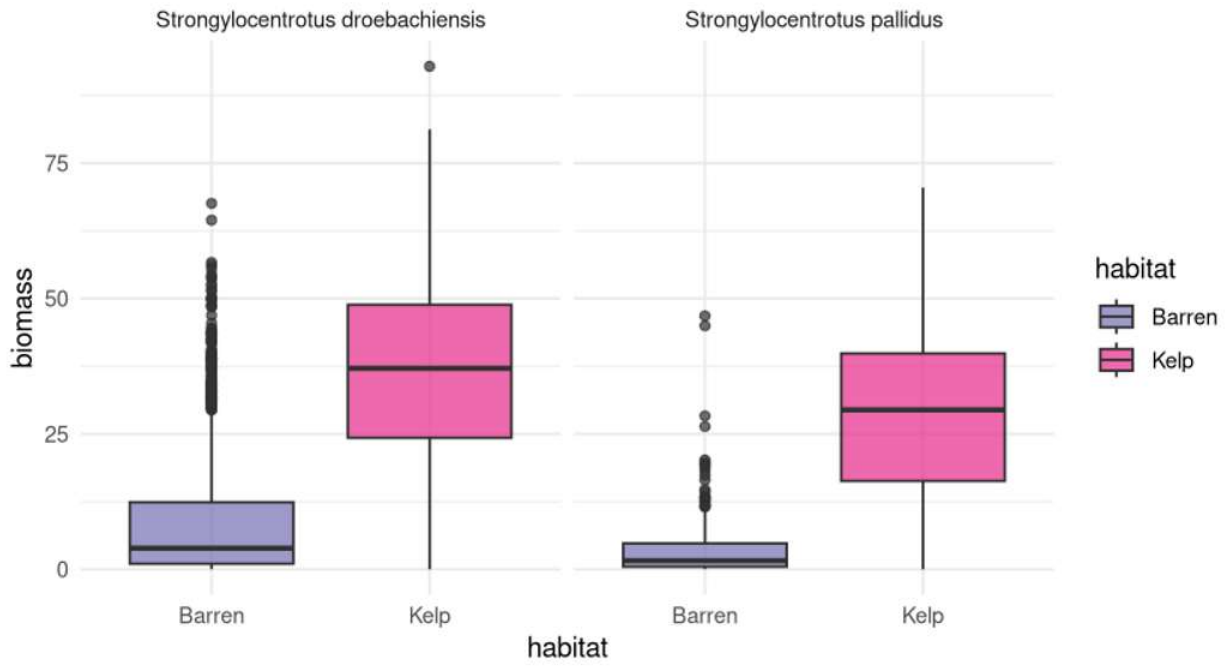


Figure 3. Distribution of biomass across habitat types (barren vs kelp), shown separately for each species. Boxplots represent the median, interquartile range, and variability of biomass. Individuals in kelp habitats consistently have higher biomass than those in barren habitats, suggesting a strong effect of resource availability.

Discussion:

This analysis confirms that sea urchin biomass is primarily driven by body size, consistent with my general biological expectations for these species. The strong relationship between diameter and biomass highlights the importance of sea urchin size in understanding their population dynamics.

Differences in scaling between species suggest that they may allocate energy differently or exhibit different growth strategies. Species differences in biomass suggest underlying biological variation between *Strongylocentrotus droebachiensis* and *Strongylocentrotus pallidus*. Although the difference is relatively small compared to the effect of size, it remains statistically significant, indicating that species identity contributes to biomass variation beyond simple size differences.

Habitat was a major factor influencing biomass, with substantially higher values observed in kelp habitats. This likely reflects differences in food availability, as kelp forests provide more food sources compared to barren habitats. The magnitude of this effect suggests that environmental conditions play a critical role in shaping sea urchin growth.

The effect of year was relatively small; this suggests that environmental conditions were most likely similar between 2022 and 2023, or that interannual variation is minor compared to the effects of size and habitat on these species. A longer time series would be necessary to assess broader temporal trends in this region and these species.

A limitation of this study is the unbalanced sampling design, particularly regarding how many more observations from barren habitats and the uneven amounts between the two species. Future work could incorporate additional environmental variables such as temperature or food availability. Additionally, the analysis does not include other environmental variables (such as temperature), which could further explain variation in biomass. Despite these limitations, the analysis provides a clear and reproducible assessment of the factors influencing sea urchin biomass.

Conclusion:

Sea urchin biomass is strongly influenced by size, with additional contributions by species and from habitat. Scaling relationships differ between species, and kelp habitats support larger individuals. These findings emphasize the importance of biological and environmental factors in shaping sea urchin growth and habitat preference and can provide a foundation for future ecological studies.

AI Use Statement:

AI tools (ChatGPT) were used to assist with code writing, organization, and choosing and processing statistical approaches on R Studio. All data interpretations, writing, code edits and comments, and readme instructions were created by me. AI was not used to generate the raw data (see above citation) nor was it used for results after data was processed.